

Paraflow: A Computational Design Tool for Support-free Multimaterial 3D Printing

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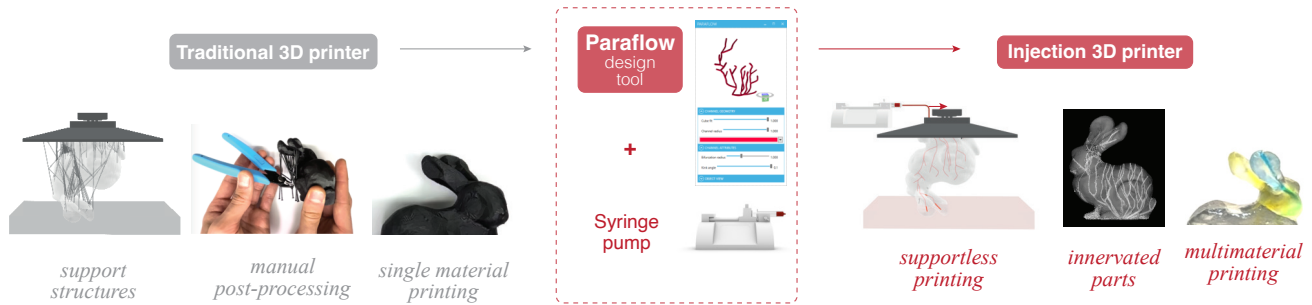


Figure 1: With minimal hardware modifications, a traditional and widely-available single material 3D printer requiring cumbersome and wasteful support structures, which must be manually post-processed by the user, can be converted to an injection 3D printer, which minimizes the need for such supports via our Paraflow design software and allows the user to print with multiple materials at once.

ABSTRACT

While 3D printing affords designers unprecedented geometric complexity, currently it requires cumbersome support structures that are not user-friendly in many ways: materially wasteful, human labor-intensive, time-consuming to remove, damaging to surface finish, and often unreliable in ensuring printability at all. We describe our novel 3D printing method, Injection 3D printing, which, as an add-on extension to existing resin printers requiring minimal hardware changes, offers an alternative to support structures. Our approach innervates the part with fluidic channels that introduce one, or multiple different, resins into the printer, effectively offsetting forces otherwise necessitating support. To allow any resin 3D printer user to implement multimaterial injection printing on their system, we present Paraflow, a computational inverse design tool that innervates a 3D model to be 3D printed by our approach.

CCS CONCEPTS

• **Applied computing** → **Computer-aided design**; • **Computing methodologies** → **Parametric curve and surface models**.

KEYWORDS

Generative design, Design for Additive Manufacturing, 3D printing

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1 INTRODUCTION

3D printing is increasingly allowing designers to rapidly fabricate highly complex computer-aided design (CAD) models unmakeable by traditional manufacturing methods, at increasingly high resolutions, increasingly large volumes, and with increasingly diverse materials. Moving beyond historic nozzle extrusion-based 3D printers, which were widely used for prototyping brittle acrylic polymers, recently UV light-based resin 3D printing in particular has opened up lattice geometries printable with elastomeric materials displaying unprecedented deformation behaviors, or even with shape changing materials. Emerging applications include interactive devices, haptic objects, and augmented reality interfaces, among others [8].

However, a remaining significant bottleneck to the more widespread use, and user-friendliness generally, of 3D printing are so-called support structures, the most popular and widespread design feature users employ to ensure 3D digital models are 3D printable. Supports, which connect the growing part to the build surface, are required for most 3D printing jobs, generated after the design stage but before printing to offset forces that otherwise preclude printing. Supports are cumbersome and not user-friendly in a number of ways, however, requiring significant manual time to cleave in post-processing and damaging the surface finish of the printed object once removed (**Figure 2**). Not recyclable after printing, supports

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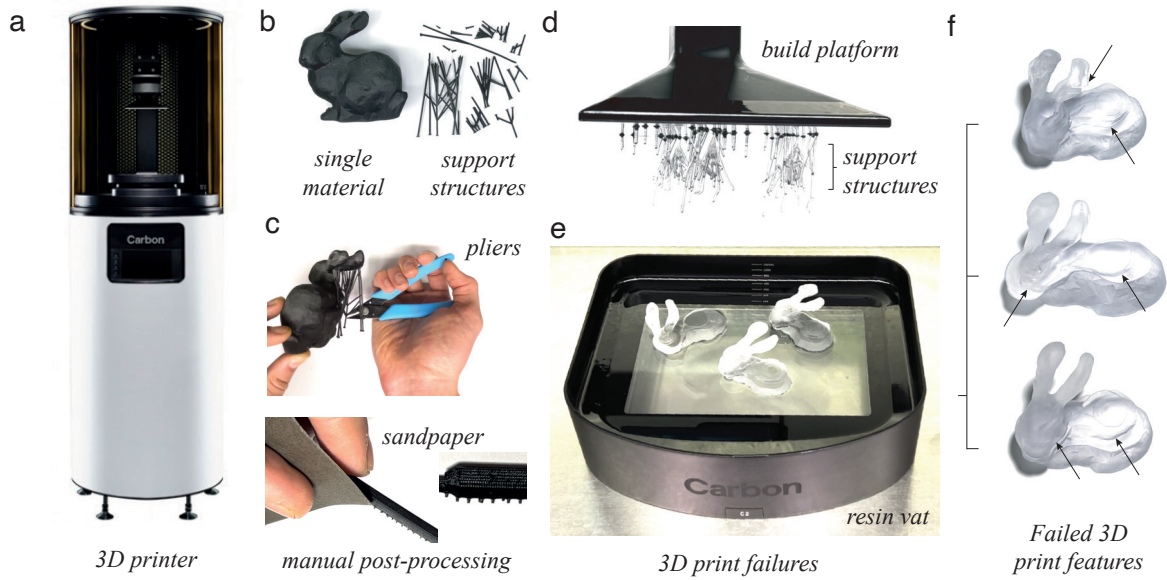


Figure 2: Current resin 3D printers (a) cannot offer designers effective multi-material print options (a), require manual post-processing of supports (c), and often fail due to high forces during printing (d-f).

also waste copious amounts of potentially expensive extra material, which is both unsustainable environmentally and financially costly to the user. While automated support structure generators are commonplace, moreover, often these do not align with a designer’s wishes, requiring significant manual design effort to revise. Finally, such supports often fail to prevent print failures in the first place, resulting in either severe part defects or whole scale failure.

Here we describe our novel 3D printing method that promises to obviate, or reduce the need for, such support structures, with the goal of improving the user-friendliness of resin 3D printing. Injection 3D printing (Figure 3) is a fast, high-resolution vat-based 3D printing method that infuses resin through fluidic channels within the 3D model itself. This has been shown to effectively offset suction forces, hence promising to reduce the need for supports, during printing [14]. In addition, such channels can rapidly and spatiotemporally infuse multiple materials into the printer vat, or the object itself, through software-designed channels specifically designed to craft multimaterial objects, either with variable deformation behaviors or sensing materials. No commercially-available resin printers can to date allow the user to perform such multimaterial printing. Injection 3D printing requires minimal changes to existing, widely available, low-cost 3D printers: any commercially-available syringe pump is the only required hardware add-on.

Injection 3D printing depends upon the careful generation of a fluidic network within an object to satisfactorily offset suction forces and guide multimaterial distributions. To that end, here we also describe our new physics-informed design tool, termed Paraflow, that allows users to implement multimaterial injection printing on their otherwise single material 3D printer, with minimal hardware changes. The main contributions claimed in this work are thus twofold:

- (1) a novel 3D printing method, Injection 3D printing, which mitigates the need for support structures in resin 3D printing; and,
- (2) an accompanying computational design tool, Paraflow, which enables designers to, with minimal hardware changes to existing printers, perform fast multimaterial resin 3D printing.

2 RELATED WORK

2.1 Computational design of 3D printed support structures

Numerous commercially-available support structure design tools for 3D printing users have been proposed and developed, both in academia and industry. All seek to minimize their volume to limit waste and detriment to surface finish, including via thin support shell enclosures [10], long-spanning bridge structures [4], branch-merging mechanisms to minimize redundancy [21], and slender tree-like structures to replace dense multi-column networks [30]. All approaches are valuable improvements, but still require supports and hence constitute mitigating strategies that fail to solve the fundamental time and material waste problem for the user; moreover, reducing support volumes often sacrifices reliability, potentially causing entirely failed prints. Optimizing the (re-)orientation of object(s) during printing can also minimize supports required [26], including taking into consideration aesthetic factors [40] or optimizing multiple object packing on the print bed [31], but a limited number of feasible part degrees of freedom or arrangements exist. Decomposing an object into several joinable parts requiring fewer supports has been proposed [15, 38], but this severely degrades mechanical properties. The lack of effective ways to mitigate the cumbersome post-processing and waste required by supports motivates the alternative described in this paper.

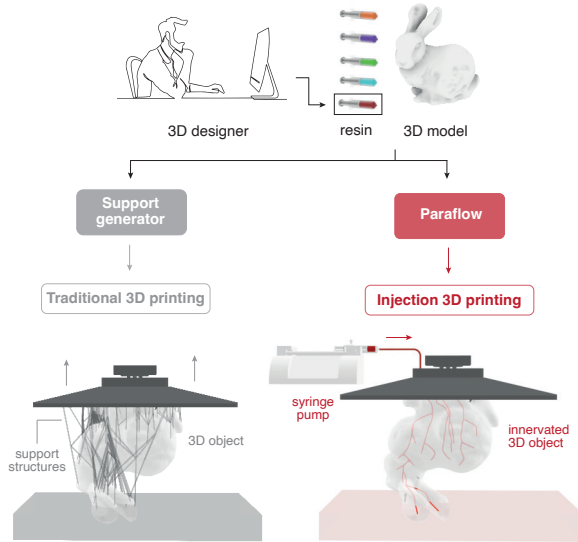


Figure 3: Traditional 3D printing uses support structures to attempt to withstand high forces during printing. To replace such supports, Injection 3D printing produces a fluidic network offsetting these forces directly to ensure a designer’s model is printable.

2.2 3D printed soft fluidic interactive devices

Beyond replacing support structures, the channels co-designed into 3D printed objects are potentially integral to the final object as well, and hence functional themselves. In this area, HCI researchers have begun actively investigating the versatility of 3D printed embedded fluidics to fabricate highly flexible, interactive devices, an area of historic interest to the HCI community, for a diversity of users e.g. students and the visually impaired [25]. Wang et al.’s “xPrint” [33] and Groeger et al.’s “Hydroprinting” use water-transfer to produce objects with embedded electronics for touch sensors [7], while Hanton et al.’s “ProtoSpray”, using spray coating for similarly interactive luminescent displays [9]. Ueno et al. innervate physical object with conductive wiring for complex touch operations [29], while Savage et al. demonstrate how internal pipes can be functionalized with inserted media [20]. Zhakypov et al.’s “FingerPrint” leverages 3D printed internal channels for embedded actuation of fingertip haptic devices [39], Bächer et al. design such channels as conductive deformation sensors [2], and Yan and Peng’s “FabHydro” use them for actuation of diverse user products [37]. Particularly relevant to our work is Sun et al.’s holistic parametric design tool to assist users in designing embedded interactive channels [27]. Distinguishing Injection 3D printing and Paraflow, however, is that no added assembly steps are required of the user after printing, instead infusing multiple materials into objects fully concurrent with printing.

2.3 3D printed soft multimaterial input devices

Emerging multimaterial 3D printing platforms represent another strategy of recent interest to the HCI community for fabricating interactive objects, ranging from wearables [5] to augmented reality devices [3]. Soft materials have been of particular interest, including

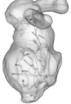
3D printing feature		Traditional	Injection
	Support structures	More required	Fewer required
	Lattice geometries	Yes	Yes
	Elastomeric materials	Yes	Yes
	Embedded deformation sensors	Multiple steps	Single step
	Multimaterial printing	No	Yes

Table 1: Summary comparison of fabrication limitations and possibilities afforded by traditional 3D printers and injection 3D printers salient to the production of interactive devices for human-computer interaction.

elastomers [23] and less traditional, but still printable, flexible materials [11, 19]. Combined with lattice architectures achievable only with 3D printing enables even more complex interactions, including for e.g. haptic gloves [17]. After printing, such multimaterial structures can be functionalized with embedded capacitive touch [22] or optoelectronic [34] sensors. To design such structures, Takahashi et al.’s parametric tool allows users to craft multimaterial filaments [28], and Vidimce et al. allow users to visually design heterogeneous material properties [32]. Benefiting from this prior work is our printing platform, which allows UV resin printer users to infuse several different materials through, for instance, deformable lattice structures. Our tool is unique, however, in its application to resin 3D printing, which is both faster and produces parts with superior resolution and surface finish than extrusion-based approaches. Resin printer users have not been able to efficiently design with multimaterials to date: proposed multimaterial resin printers employ vat-switching methods, e.g. a rotating carousels [12], which are hardware-intensive, expensive, and significantly slow down printing. With Injection 3D printing, high resolution channels rapidly infuse different resins into the part spatioselectively, without adding extra process steps.

3 INJECTION 3D PRINTING

Injection 3D printing is implemented as an add-on extension to other vat-based resin 3D printers, a class of 3D printers rapidly gaining popularity among users in a variety of industries thanks to their high resolution, speed, and exceptional surface finish [35]. In such printers, a vat of liquid resin is spatioselectively solidified by a digital light projector beneath the vat, and the object is attached to a build platform that ascends as the object is steadily created [6]. Traditional resin printing is largely single material and require extensive supports (Figure 3), as material cannot readily flow in the

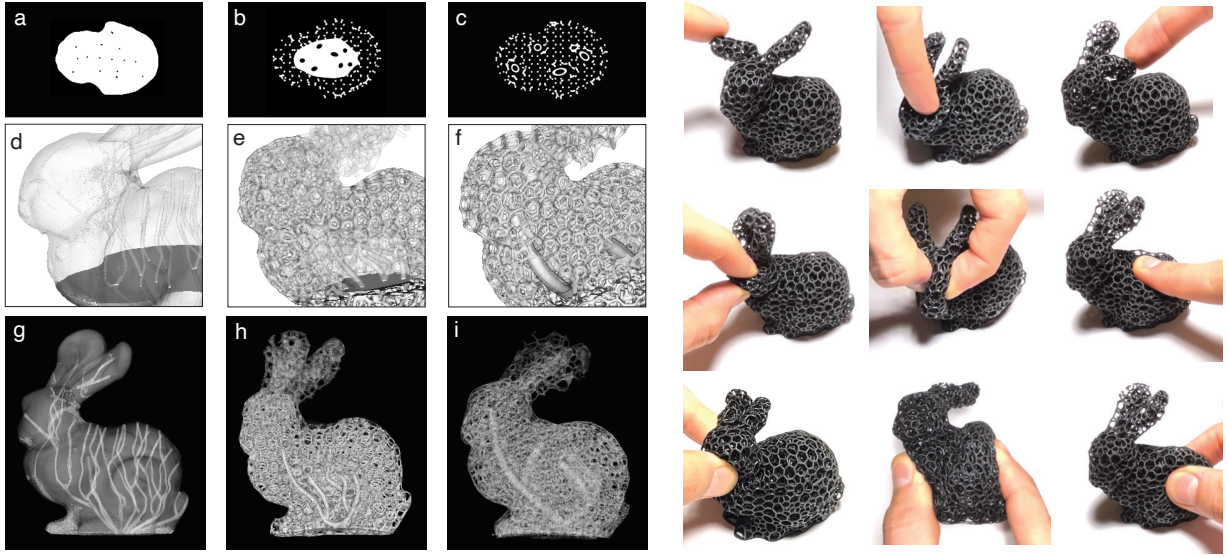


Figure 4: Paraflow can innervate fully solid bodies (a,d,g), heterogeneous solid-lattice bodies (b,e,h), or fully latticed bodies (c,f,i) capable of complex and reversible deformations. At top is shown example image slices for printing, at middle isometric CT scans of print results, and at bottom CT scans of full print results.

vat unaided. In Injection 3D printing, resin is actively introduced through the build platform, through channels software-engineered into the part, finally into the vat.

Like other such vat-based approaches, Injection 3D printing allows users to print intricate lattice structures with materials such as elastomers, for instance for highly deformable objects. However, it improves the user flexibility of vat-based printing in a number of ways, which are summarized in **Table 1**. If a user wishes to print solid bodies, an extensive network of support structures is typically required; by our method, these are replaced with a variable density fluidic network (**Figure 4a**). Experimental validation that injection can alleviate suction forces that otherwise require supports may be found in the Appendix. While printing, such channels can also be infused with sensing materials for deformation monitoring. Channels designed for Injection 3D printing can also innervate lattices for e.g. interactive devices, which can be especially intricate when made with vat-based processes (**Figure 4c**). In reality, parts are often heterogeneous combinations of latticed and unlatticed regions (**Figure 4b**).

3.1 Injection 3D printing user workflow

To implement Injection 3D printing, either low-cost home built printers or commercially available printers can be modified. For custom printers, traditional printer hardware elements, namely the build platform and light projector, can be coordinated with the add-on syringe pump by an inexpensive Arduino MEGA 2560 microcontroller mounted with a RAMPs 1.4 shield running open-source Marlin firmware. Any programmable syringe pump can be used to inject resin; we specifically used a Harvard Apparatus PHD Ultra syringe pump to controllably administer resin flow during and/or after printing. Our own custom user interface allowing the

designer to adjust resin flow during or after printing as desired. As an extensible add-on, users do not need to learn the constraints of an entirely new printing method to implement Injection printing; rather, they can modify existing and well-established resin 3D printing workflows with our new add-on.

4 PARAFLOW DESIGN TOOL

To aid users in implementing an injection printing add-on with their existing printers, we present a computational design tool for innervating injection channels in a 3D model. We call our tool Paraflow, which we summarize below.

4.1 Formulation of the Design Problem

Our goal is, for an arbitrary 3D model provided as input by a CAD modeler and 3D printer user, to construct a fluidic network that offsets suction, in order to effectively replace, or minimize, support structures required. To generate such a suitable 3D printable network, we formulate the design problem as a path planning optimization problem where, for an arbitrary 3D geometry to be 3D printed via a set of 2D slice layers, we computationally design a corresponding fluidic network that innervates the part to sufficiently distribute one, or multiple, materials during printing. The goal of such a network is to offset suction forces at all layers, which we also flag to the user. While traditional support structure generators also automate such printability analysis, such that the CAD modeler and 3D printer user does not have to intuitively guess when print failures will occur, instead of subsequently imposing cumbersome extra solid material to accommodate such forces, we design a fluidic network to instead offset suction directly while fabricating the user's part.

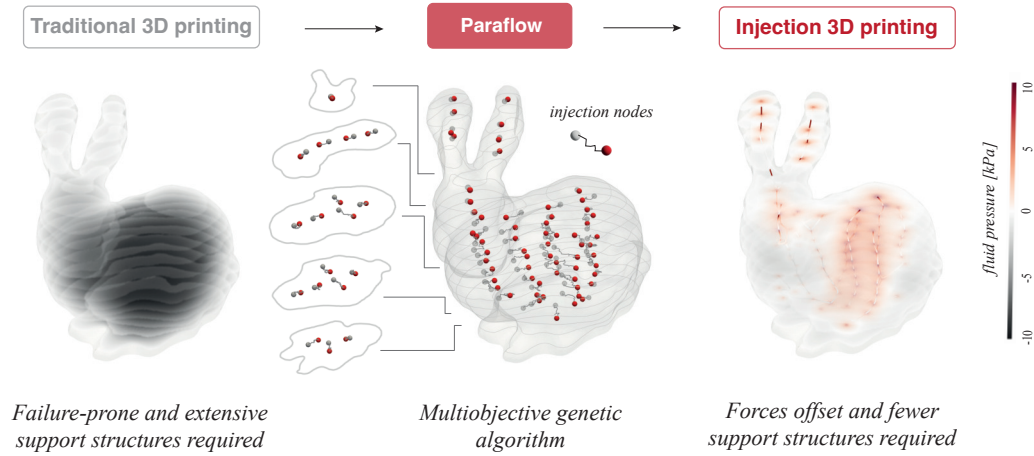


Figure 5: Paraflow generates a fluidic network that, by facilitating injection during printing, offsets the high negative pressure gradients during printing that otherwise require support structures to avoid failure. Above, dark grey regions flag to the user areas of the to-be-printed part at risk of suction-related failure (left), which via fluid injection at optimal locations as determined by a multiobjective genetic algorithm (middle) are offset with high-fluid pressure regions instead (right).

4.2 Definition of the Design Space

The 3D model inputted by the user defines the computational bounding volume for the generation of the 3D fluidic network, which must be monotonically connected to a single, or optionally multiple, input injection sources at the build platform. While many generative procedural modeling tools for branching structures have been developed in the computer graphics community, often inspired by biological growth mechanisms [13], it is important that those generated for Injection 3D printing are indeed easily 3D printable, and hence designer-friendly. Considerations such as the challenges to the 3D printer user associated with flowing material through, or flushing material out, of channels, clogging of channels that are too narrow during printing, and collapse or warping of free-standing channels could all preclude the physical realization of an arbitrary digital fluidic design. **Figure 4** above confirms that, indeed, Paraflow produces intricate yet still 3D printable fluidic networks: the design of a (comparatively dense) fluidic network, for the case of the Stanford bunny, while highly complex, is printable, thanks to taking into account the limitations of the printing process. Bifurcations leading from one to two, three, or many branches are achievable. Vessels with good dimensional accuracy without over-cure that facilitate post-cleaning are also achievable, highlighting the design flexibility for the user.

4.3 Computational Design Methodology and User Interface

Our inverse design approach (**Figure 5**) originates a fluidic network with a single input injection point connected to the build platform and, thus, the syringe pump. Then, injection channels are positioned within the user-provided model at optimal positions as determined by surrogate fluid dynamics modeling, in order to offset suction during printing to minimize need for supports. As it grows in complexity, the network is modeled as an electrical

circuit, as is typical for pressure-driven fluidics [18]. Branches may be either open (with outlets), or closed (without outlets), thus not contributing to fluid flow. The final network, encoded as a graph connectivity matrix of nodes and edges, is converted to a series of smooth NURBS curves, swept to produce a positive network 3D geometry, and finally Boolean differenced with the original B-rep CAD model via surface-to-surface intersection [24] to produce negative channels, hence an innervated part. Details of our algorithm may be found in the Appendix.

Rather than presenting the user with a single generatively designed fluidic network, Paraflow presents multiple options for innervating networks to the user, some of which may be more desirable for functional and/or aesthetic requirements. While there is one optimal network configuration that minimizes suction during printing, many possible configurations satisfy the design space constraints described above, with higher variance generally in part footprints with higher cross sectional area. Some evolved designs, however, may be more satisfactory for the given user preferences. In this way, Paraflow resembles increasingly popular generative design software tools for mechanical property optimization, academic [36] or commercial [1], that, unlike more traditional topology optimization software, present users with a range of potential viable designs, from which the user can select a single option. All fluidic networks Paraflow offers the designer are viable for a given printer and material configuration provided by the designer, with several example solution spaces shown in **Figure 6**. If the user requires multimaterial printing, the flow modeling integrated into Paraflow also instructs the user as to what flow rates to administer on their injection 3D printer to achieve a desired fluid distribution within the vat, for all slices at all times during the 3D print (**Figure 7**). This is particularly important for multimaterial printing, where it is desirable to pattern the vat with different fluids that will, ultimately, comprise the printed object.

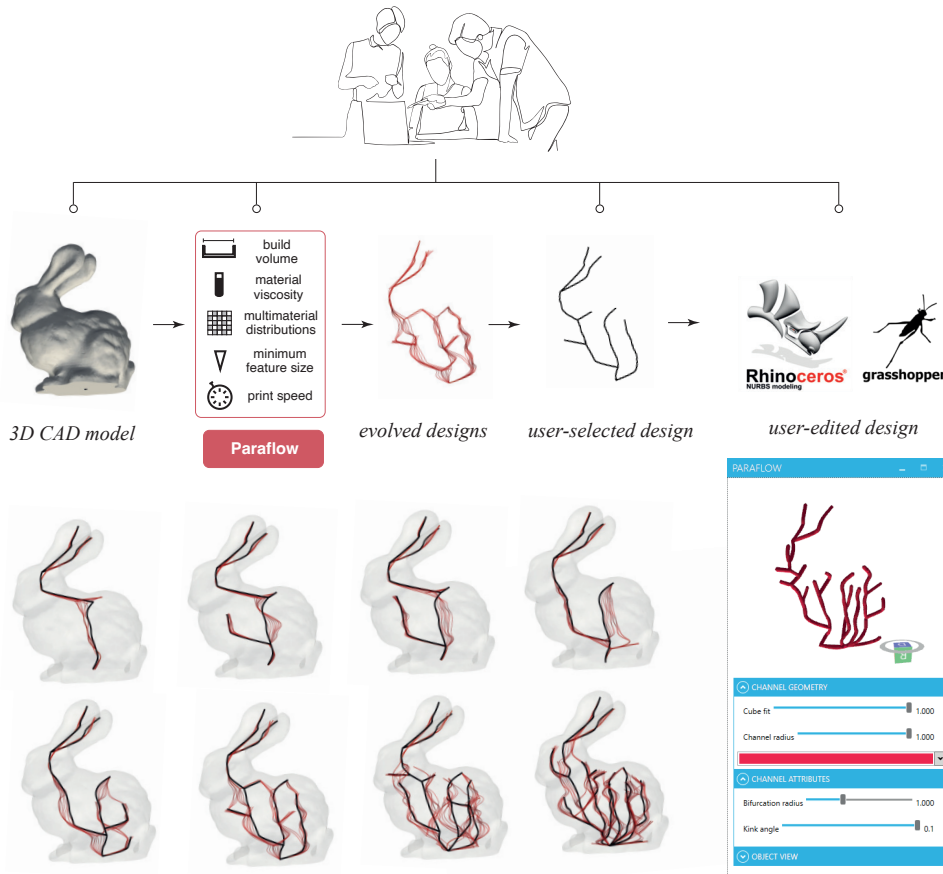


Figure 6: Paraflow offers the designer a range of viable fluidic networks innervating an input model, from which a desired configuration is chosen for injection printing. The user is involved in the generative tool for (1) providing the initial CAD model, (2) specifying resin 3D printer model and material, (3) choosing an evolved design, and (4) adjusting the design considering product requirements.

The user can continue to edit any of the generatively designed networks, since rather than outputted as an uneditable mesh directly for 3D printing (in .stl format), a fluidic network geometry generated by Paraflow is readily loadable into parametric design software, specifically Grasshopper for Rhino3D [16], where the designer can continue to spatially adjust control points describing NURBS curves that comprise the network, if desired. We design a simple user interface to expose such post-processing parameters to the designer, who can thereby interact with the generative design tool. Particularly for sharp corners or detailed features in a model, there may be reasons for the user to wish to adjust the generatively designed fluidic network. In addition to changing the channel geometries themselves, designers may customize high-level network parameters such as the curve geometry used to produce the network (i.e. NURBS, interpolated splines, control point splines, or polylines) along with channel radius, curvature, and bifurcation radius. To automate the visualization process for the user, provided in the Supplementary Materials is a helper GHPython data post-processing helper script for Grasshopper.

5 DISCUSSION AND LIMITATIONS

Experimental validation that our design tool successfully offsets suction, so as to alleviate the need for supports during printing, can be found in Appendix Section 1.3. Whereas traditional 3D printing

without injection experiences high suction forces, as evidenced by a load cell mounted on the printer, injection through a network with three branch termini into the vat fully reverses pressure forces during printing. This is the mechanism by which our tool partially, or fully, offsets suction to reduce reliance on supports. However, there are still several limitations of our approach and design logic. First, the length scales of the fluidic network are limited by the resolution of the printing method, both the smallest positive and negative feature resolvable. Typical channel sizes for injection 3D printing are $350\ \mu\text{m}$ in radius. Second, so that disruption to part mechanical properties is minimized, our workflow typically post-cures (solidifies) resin trapped within channels, which may require a thermal initiator to be added to the resin. Finally, we emphasize that not all product geometries may allow for the complete removal of supports by our method. In particular, supports are also needed for printing so-called "island" geometries that would otherwise have no pre-existing part onto which to attach. In this case, supports will still be necessary. Nonetheless, in cases where large part cross sections threaten print failures, our design tool and preliminary validation results suggest fluidic inverse designs as generated by our design tool provide a promising alternative.

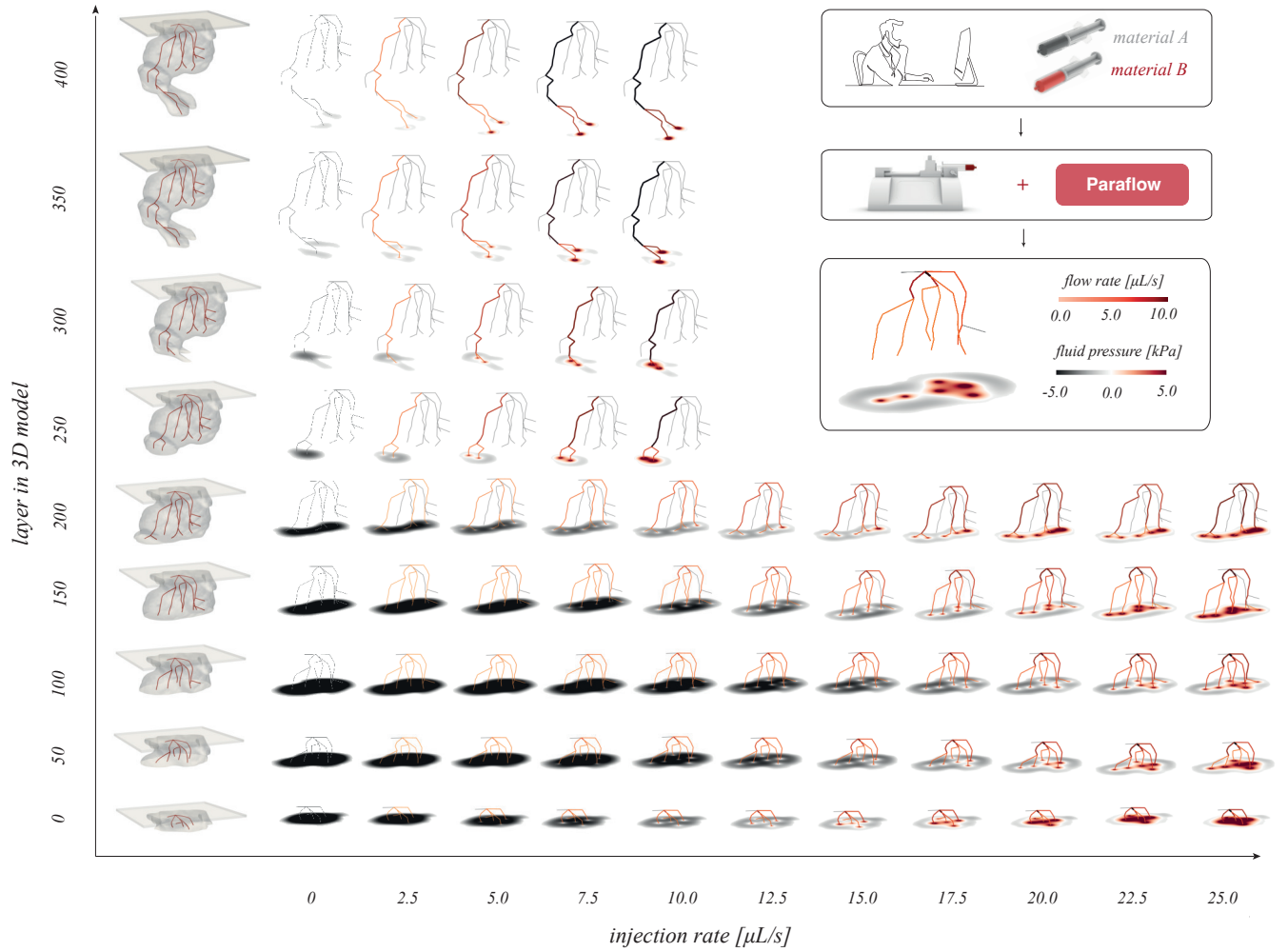


Figure 7: Paraflow, having generated a fluidic network before printing, enables the designer to programmatically control the position of border regions between fluids, and hence multimaterial boundaries, during Injection 3D printing. Shown above are such simulations for varying slices in the 3D model, for varying example input injection rates chosen by the designer.

6 CONCLUSIONS AND FUTURE DIRECTIONS

Here, we presented our ongoing work on a novel 3D printing method, Injection 3D printing, and its accompanying 3D generative design tool, Paraflow, which allows users to minimize the need for supports on their existing resin 3D printers, and also engage with multimaterial resin 3D printing. Our method, with minimal changes to the hardware of existing resin printers, produces printable fluidic networks that reduce forces, and the need for supports, during printing and enable the potential for fast multimaterial printing. Future applications-oriented work will further explore use cases of this extensible add-on to commercial resin 3D printers in the area of highly interactive input devices, leveraging multimaterial latticed elastomers. Future technical work will focus upon validating what geometries can be printed with fewer or no supports with our tool,

developing further support for fluid injection through multimaterial lattices, and integrating real-time vision control into injection 3D printers to allow users to better control multimaterial flows in real-time.

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